

Master Plan

Master Dashboard - Live Page Builder CRM - Quotations & Invoices - Go-Live

Full-stack implementation plan based on the completed deep audit,
the approved design mockup and the five locked decisions.

87-page website - No-code block builder - Lead-gen CRM with live chat & WhatsApp - Server-PDF
quotation pipeline - Vercel hosting.

Status	FOR APPROVAL - development starts after sign-off
Version	1.0 - 28 September 2026
Scope	Phases 0-6 - 13 weeks
Hosting	Vercel (site + admin + API) - Hostinger = domain only
Based on	AUDIT-REPORT-2026-09-28.md + design mockup + admin screenshots

WOODEX - Backend Master Plan

Version: 1.0 - **Date:** 2026-09-29 - **Status:** awaiting approval (phase gates) **Relationship:** extends MASTER-PLAN.md (Phases 0-6). Phase 0, 1, 2 are delivered; this plan governs the **backend/module build-out** that follows, in build order.

0. Current State (what is live vs planned)

Module	State
Dashboard home (KPIs, analytics, widgets)	? live
Page Builder + typed blocks (87 pages converted)	? live
Save draft -> Publish -> GitHub -> live	? live (tested end-to-end)
Enquiries (CRM list)	? live
Reports (range, KPIs, chart, CSV)	? live (fills after Supabase migration)
Pages management (add/import/status/SEO)	? B1
Live Chat, Header/Footer/Global Sections, Galleries, Image Optimization, Schema, WhatsApp module, Integrations, Reports automation	? placeholders (Coming-soon view)
Runtime backup import/export	? B3
AI/provider integrations in Settings	? B4

Pages vs Visual Builder (today): both open the same combined editor screen. **B1 splits them:** *Pages* = management of all 87 pages (list, add, import, status, SEO, duplicate, delete-with-redirect); *Visual Builder* = the section editor.

Decisions required (approval gate 0)

- **D6 - WhatsApp number.** Site currently uses **+92 322 4000768** (wa.me/923224000768, 230 occurrences). Requested number was **+92 322 4200768**. Confirm which is correct before any global replace (one digit differs: 4000768 vs 4200768).
- **D7 - WhatsApp scope.** Original D2 was "wa.me only". B2 proposes adding Meta Cloud API for **outbound notifications** (new enquiry -> WhatsApp alert to admin, quotation sent -> client ping). wa.me remains the customer-facing click-to-chat.
- **D8 - AI integrations.** B4 connectors: **Claude (Anthropic)**, **Codex/OpenAI**, **Hermes** - used for content drafts, meta descriptions, reply suggestions. Confirm providers + that API keys are customer-supplied (stored server-side only).

B1 - Page Management System (87 pages) - *priority*

Goal: Pages becomes the control room for every page; Visual Builder is the editor.

Deliverables

1. **Pages list view** (view-pages, cms-pages?action=list v2)
 - All 87 pages: title, slug, type (service/location/insight/project/...), status (Published / Draft-changes-pending), SEO score (title+desc+canonical+OG present), sections count, last published (GitHub commit time), open-editor / view-live links.

- Search box + status filter + type filter; sortable columns.
2. **Add Page wizard** (blueprint Section 4): Step 1 type -> Step 2 template donor -> Step 3 content (title/slug/hero placeholder) -> Step 4 SEO -> Step 5 Preview -> Publish. Backend: cms-pages?action=create (exists) extended with template presets.
 3. **Page import system**
 - cms-pages?action=import&slug= - re-split a live page into blocks+meta (promotes tools/blocks-convert.py logic to a server module _page-split.mjs shared with the offline tool; identity validation required before overwrite).
 - Import external HTML: paste/upload HTML -> validate (main-element required, scripts allowed but sandboxed to same-origin assets) -> register page.
 - "Re-sync from live" per page (discard local drafts, rebuild from current HTML).
 4. **Page actions:** duplicate (new slug + donor copy), archive (hide from sitemap, keep file), delete with redirect (exists - surfaced in list UI).
 5. **Visual Builder entry:** editor deep-link ?page=/slug/ opens straight into the builder; sidebar Visual Builder = "resume last edited".

API

GET list|get - POST save|publish|create|import|resync|duplicate|archive|delete

Acceptance gate (B1)

- 87 pages listed with correct status/SEO columns; search+filters work.
- Wizard creates a page end-to-end (draft -> preview -> publish -> live URL).
- Import of a live page reproduces it byte-identical (identity check) before any edit.
- Duplicate produces an independent page; delete inserts redirect row.

B2 - WhatsApp API (number per D6/D7)

Deliverables

1. **Settings -> WhatsApp:** number, greeting text, off-hours message, admin notify toggle. Single source of truth (blueprint Section 23) - header, footer, floating button, contact page all read it (replace the 230 hardcoded occurrences with assets/js/site-config.js values, published via cms-settings).
2. **wa.me click-to-chat** everywhere (already present; re-pointed from settings).
3. **Cloud API outbound (if D7 approved):** netlify/functions/whatsapp-send.mjs - server-side Graph API call (token env WHATSAPP_TOKEN, phone id env), used by: new-enquiry admin alert, quotation-approved notification. Template messages only (Meta policy), rate-limit + idempotency key.
4. **Inbound (phase 2 of B2):** webhook endpoint whatsapp-webhook.mjs -> creates/updates enquiry in CRM with source=whatsapp. Verify HMAC (X-Hub-Signature-256).

Acceptance gate (B2)

- Number editable in Settings -> updates all site touchpoints after publish.
- Test alert: create enquiry -> admin WhatsApp received (or toggle off gracefully).
- Webhook (if enabled) creates a CRM row; invalid signatures rejected.

B3 - Runtime Backup / Import / Export

Goal: never depend on manual exports; any point-in-time restore.

Deliverables

1. **Export** (cms-backup?action=export&scope=all|cms|content|media)
 - JSON dump of all CMS tables (Supabase REST), content/pages/ (**drafts**), **vercel.json redirects**, **settings row**; **media** manifest (**URLs+metadata**).
 - Streams as single .json (or zip when media binary included) download.
2. **Scheduled GitHub backup**: nightly GitHub Action-free design - a backup job in cms-backup?action=snapshot committing backups/YYYY-MM-DD.json to a backups branch (keeps 14 days, oldest pruned).
3. **Import / restore** (cms-backup?action=import)
 - Upload JSON -> schema validation -> **dry-run diff preview** (what would change) -> confirmed apply; every import writes an audit-trail entry.
4. UI: improve existing **Backups** view - snapshot list, Download, Restore, Export-now, Last-backup status on System Health.

Acceptance gate (B3)

- Export downloads a valid JSON that passes import dry-run with 0 errors.
- Snapshot branch shows dated commits; retention prunes at 14.
- Restore of a dry-run-tested file reproduces CMS rows (tested on a sandbox page).

B4 - Integrations Hub (Settings -> Integrations)

Goal: one place to connect AI + business services; keys server-side only.

Deliverables

1. **Integration model** (table integrations): provider, name, api_key_enc, config jsonb, enabled, updated_by, updated_at. Key write-only from UI (never returned; masked ----1234 display), service-role only on server.
2. **Providers (initial)**:
 - **Claude (Anthropic)** - content drafts, FAQ generation, reply suggestions.
 - **Codex (OpenAI)** - same surfaces, alternate provider.
 - **Hermes** - same surfaces.
 - **Resend (D5)** - quotation/invoice email sending (already planned Phase 4).
 - **GitHub / Vercel / Supabase** - status cards (read-only health, Section 27).
3. **AI endpoints**: cms-ai?action=generate {surface: meta|blog-outline|faq|reply} -> provider-agnostic adapter (_ai.mjs) with provider selected from enabled integrations; prompt templates stored in settings; output lands in editor fields (never auto-published).
4. **Webhooks out**: integrations config supports generic webhook URL + event list (enquiry.created, quotation.sent, page.published) -> HMAC-signed POST.
5. **Connection test button** per provider (real ping, clear error).

Acceptance gate (B4)

- Save key -> test -> green; key never visible again in any response.
- One AI surface (meta description) generates into the Pages SEO form end-to-end.
- Webhook fires on enquiry.created with valid signature (verified by test receiver).

B5 - Blueprint modules (build order)

#	Module	Notes
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#	Module	Notes
B5.1	Live Chat (D2)	Supabase Realtime: visitor widget (site) + admin inbox (dashboard), offline -> WhatsApp fallback. New: chat_channels/messages tables (add to migration), cms-chat.mjs.
B5.2	Header & Navigation builder	Controls per blueprint Section 8; writes header include used by publish pipeline.
B5.3	Footer editor	Same pattern; global footer section.
B5.4	Global Sections	Reusable CTA/banner sections referenced by pages (block type global_ref, resolved at publish).
B5.5	Galleries + Image Optimization	Media folders/albums; sharp/AVIF pipeline via cms-media extension + nightly optimize job (ties to Phase 0 image work).
B5.6	Schema manager	Per-page JSON-LD types (FAQ, Service, LocalBusiness, Article) rendered into <head> at publish; validation on save.
B5.7	Activity	Activity feed view from audit_trail (Section 28) with filters.

Each module = its own approval gate (build -> preview -> "done ?" -> next).

B6 - Backend hardening (continuous, parallel)

- API envelope:** consistent {ok|error, data} + machine error codes; retrofit new modules first, legacy endpoints on touch.
- Rate limiting:** IP+route buckets for public endpoints (enquiry-submit, analytics-track, whatsapp-webhook) - in-memory now, Supabase-backed later.
- Audit coverage:** every mutation in new modules writes audit_trail (who, what, diff summary).
- Secrets hygiene:** all keys in Vercel env or integrations table - never in repo; rotation checklist (exposed chat credentials from setup are on it - rotate Supabase + Vercel + GitHub PAT after go-live).
- CI smoke suite:** script tools/smoke.mjs - syntax checks, 86-page render identity, block identity, nav/TITLE integrity (the validators used in Phases 1-2) runnable in one command; wire to a GitHub Action on push.
- Health enrichment:** System Health adds CMS API, Media, Publishing, Migrations rows (Section 27) fed by cms-health real checks.
- Sitemap automation:** regenerated on every publish (add to publish pipeline).

Suggested build order & gates

```
D6/D7/D8 confirm --? B1 Pages management --? B3 Backup/Export --? B4 Integrations
| ^
+--? B2 WhatsApp (D7) -----+
B5.1 Live Chat (can run parallel with B1) |
B5.2-B5.7 after B1 (all touch publish pipeline) B6 continuous
```

- **Gate rule (unchanged):** each B-phase ends with your test on the preview +

"done ?" before the next starts. Deployment to Vercel stays the final step after your overall approval.

Go-live linkage (from MASTER-PLAN)

Renames/keeps: media metadata columns, project publishing columns, cms_users, role testing, test publish, settings publish, desktop/mobile dashboard test, public page tests, **credential rotation**, final acceptance - executed with B3 (backup safety net in place first) immediately before cutover.